Ali Tarek Mohamed

alitarek115.at@gmail.com | +201152526996 | Giza, Egypt | linkedin.com/in/atarek100 | github.com/Ali-Tarek | alitarek.vercel.app

PROFILE

I'm a software engineer with a passion for building scalable and efficient full-stack applications. I specialize in both frontend and backend development, primarily using React and Express. I'm quick to learn new technologies and adapt to different frameworks and tools. My experience includes building clean, responsive user interfaces, designing normalized database schemas, and implementing secure REST APIs

EDUCATION

Bachelor's degree in computer science - Information Systems department Cairo University 09/2021 – 07/2025

- **Relevant Courses:** Data Structures, Database Systems, Software Engineering, Operating Systems, OOP, Computer Networks, Algorithms Design and Analysis

PROFESSIONAL EXPERIENCE

Scale Ai - LLM Engineer 03/2024 – 10/2024

- Designed and implemented prompts to assess large language models (LLMs) for code generation and multi-step reasoning.
- Evaluated model outputs through structured rubrics, pinpointing reasoning and linguistic errors.

PROJECTS

Problem Solving Assistant (Graduation Project — A+) • GitHub

- **Demo**
- Built a cross-platform assistant for competitive programmers and CS students to enhance their coding skills, featuring personalized learning roadmaps, AI-generated problem solutions, and smart debugging.
- Integrated with Code forces to track user progress and power adaptive next-step suggestions based on individual skill level.
- Developed the frontend with **React**, **TypeScript**, and **Flutter** (mobile), and the backend with **Django**, **PostgreSQL**, and **MongoDB**.
- Leveraged Hugging Face models for AI-driven features and used Selenium for automated data scraping/tracking.
- Designed the UI/UX in **Figma** and built **Power BI** dashboards for analytics and progress visualization.

ShopSphere (MERN Stack) • GitHub

- **Live Demo**
- Built a scalable, full-stack e-commerce platform with a customer-facing storefront, admin dashboard, and RESTful API.
- Developed secure user authentication, product management, and order processing using **Node.js**, **Express**, and **MongoDB**.
- Designed responsive, modern UIs for both client and admin portals using **React** and **Tailwind CSS**.
- Integrated Cloudinary for image uploads and implemented JWT-based authentication and role-based access control.

Student Management System (C++) • GitHub

- Developed a streamlined student information management system using OOP principles.
- Utilized advanced data structures for efficient data handling:
  - Binary Search Tree (BST): Efficiently searched student IDs.
  - AVL Tree: Maintained balanced height for consistent operation time complexity.
  - Min Heap: Prioritized students with the lowest GPA.
  - Max Heap: Optimized access to students with the highest GPA.
- Enhanced data retrieval efficiency and accuracy, ensuring optimal system performance.

SKILLS

Frontend Technologies: ReactJS | NextJS | Tailwind CSS | HTML | CSS | JavaScript | TypeScript

Backend Technologies: NodeJs | ExpressJs | PostgreSQL | MongoDB

Other Tools and Knowledges: C++ | Git | GitHub | REST API | Design Patterns | SOLID Principles | Problem Solving

Familiar With: Java | Python | Docker | RabbitMq | Supabase | GraphQL | Redis | gRPC

ACHIEVEMENTS

Meta Hacker Cup Contestant 2023

- Qualified through all three rounds, achieving top 8% in the Practice Round **(948/12,138)**, top 20% in Round 1 **(3,941/20,324)**, and top 57% in Round 2 **(3,532/6,193)**.

ECPC

- Participated twice in ECPC qualifications, contributing significantly to competitive programming.

LANGUAGES

English — Fluent

Arabic — Native/Bilingual